

Sohan Sharad Pillai

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Profile

Proactive first-year Computer Systems Engineering student with hands-on internship experiences. A highly motivated and analytical programmer with a diverse portfolio of projects spanning full-stack, mobile, low-level development, networking, A.I., and algorithmic optimization. Eager to apply and learn to handle complex challenges in a research-focused environment.

Education

University of Warwick, BEng in Computer Systems Engineering Sep 2025 – Jun 2028

- **Grade:** Predicted First
- **Relevant Coursework:** Sets and Proofs, Computer Organization and Architecture, Programming for Computer Scientists, Engineering Mathematics, Design of Information Structures, Systems Modelling, Simulation and Computation

The King's School, Kollam, Cambridge International A-Levels Mar 2023 – Jun 2025

- **Grade:** A, A, A
- **Subjects:** Physics, Chemistry, Mathematics
- **Activities:** School Head Boy, Executive Committee, Student Council, PCAP Python Certifications, Basketball Team, Olympiad winner (x3)

Experience

Application Developer Intern, Xploro Technologies – New Delhi, India Apr 2023 – May 2023

- Engineered a cross-platform crypto portfolio application using React Native, implementing real-time data fetching at scalable quantities.
- Collaborated with senior engineers to write production-level code while upholding industry standards.

Data Analyst Intern, Hie Capital Technologies – Mumbai, Maharashtra, India Aug 2024 – Sep 2024

- Developed automated data processing pipelines for large-scale oceanographic datasets using Python, reducing manual data input and analysis by 60%.
- Visualized complex oceanographic datasets, translating telemetry into actionable client reports.

Research & Publications

Genetic Algorithms and Their Effectiveness in NP-Complete Problems: Sudoku Dec 2024

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10.5281/zenodo.17518227

- Implemented Genetic Algorithms in Python to solve NP-Complete Sudoku grids, benchmarking convergence rates against other common algorithms

Projects

Game Companion Discord Bot 2020 – 2024

- Architected a high-throughput Discord bot serving 50,000+ users across 1,000 servers. Designed a scalable backend with Python and SQLite to handle concurrent data ingestion of 1M+ records, optimizing query latency for real-time game stat retrieval.
- Tools Used: Python, Wireshark, SQLite
- GitHub repository (private; viewable upon request)

GBA Rom Hacking with Assembly 2023

- Reverse-engineered Game Boy Advance binaries to implement custom game logic. Wrote optimized ARM Assembly and some C routines for the GBA hardware constraints, managing manual memory allocation and instruction cycle timing.
- Tools Used: C, GBA ASM

AI-Powered A-Level Exam Search Engine 2024

- Prototyped a search engine to sort and query past exam paper questions from 2015 to 2024 using Llama and Elasticsearch, for easy access to answers for enabling more productive revision.
- Tools Used: Llama, Elasticsearch, Python, Javascript, React

Technologies

Languages: C++ , C, Java, SQL, JavaScript, Python, Rust, Assembly, Bash

Technologies: React, SQLite, Pandas, NumPy, OpenCV, NodeJS, QT, Linux, Git, Wireshark, React Native, Flutter, Electron